

3-Mark Questions & Answers — Data Structures Module 3: Linked Lists

Q1. Explain insertion at beginning, middle, and end in a Singly Linked List.

1. **Insert at Beginning** ($O(1)$): `newNode->next = head; head = newNode;`
2. **Insert at End** ($O(N)$): Traverse to last node `temp->next == NULL`, then `temp->next = newNode;`
3. **Insert after Position** ($O(N)$): `newNode->next = temp->next; temp->next = newNode;`

Q2. Explain deletion of a node in a Doubly Linked List.

To delete node `del` :

```
if (del->prev != NULL) del->prev->next = del->next;
else head = del->next;
if (del->next != NULL) del->next->prev = del->prev;
free(del);
```

Q3. Explain Polynomial Representation using Linked Lists.

Each polynomial term $c \times x^e$ is represented as a node containing:

- `coeff` (Coefficient c)
- `expo` (Exponent e)
- `next` (Pointer to next term in descending exponent order)